LAB REPORT 实验报告

Lab Title	Data Types and Variables			Lab No.	01
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Lab description/objectives:

Description:

Modify example to convert a Celsius temperature into its Fahrenheit equivalent.

By using the formula, write a C program to calculate.

Objectives:

By completing the tasks to understand the using of data types and variables. From the first task, we could better choose the appropriate data types and understand the output of double type. From the second task, we could better use the arithmetic method to ensure the accuracy of the result.

Source code:

Task 1:

```
#include <stdio.h>
int main()
{
    double celsius = 40.5;
    double fahrenheit ; /* declaration and initialization */

    fahrenheit = (celsius * 9.0 / 5.0) + 32.0;
    printf("The Fahrenheit equivalent of %5.2f degrees Celsius\n",
           celsius);
    printf("    is %5.2f degrees\n", fahrenheit);

    return 0;
}
```

Task 2:

```
#include <stdio.h>
int main(){
    int upper = 20;
```

```
    int lower = 9;
    int sum = 0;
    sum = (upper*(upper+1)/2)*(upper*(upper+1)/2)-(lower*(lower+1)/2)*(lower*(lower+1)/2);
    printf("The sum of cubes from 10 to 20 is %d", sum);
    return 0;
}
```

Program outputs:

Task 1:

```
The Fahrenheit equivalent of 40.50 degrees Celsius
   is 104.90 degrees
```

Task 2:

```
The sum of cubes from 10 to 20 is 42075
```

Discussion:

1. **Most difficult parts**

   Clearly realize the relationship between 2 degrees, use correct expression to calculate.

2. **Bugs and/or Errors**

   Might forgot to use float number in the expression, finally caused the program had a wrong output.